

Practical 2

Time series features: Exploring time series II

Getting started

At the end of this practical you should be able to:

- Fit linear and polynomial trends and seasonal terms to time series data using regression.
- Use the `acf` function to study the properties of residuals.
- Understand the potential effect of smoothing and trend removal on the error structure of a series.

Before getting started, you may find it beneficial to read these short introductions to some of the packages we will use:

- [Introduction to fable](#)
 - `fable` 0.4.1 (O'Hara-Wild, Hyndman, and Wang 2024)

The Mauna Loa Data

The Mauna Loa data is a time series of 468 monthly observations of atmospheric carbon dioxide measured at the Mauna Loa observatory in Hawaii from January 1958 to December 2003. It is a built-in data set in R called `co2` as a `ts` object.

Convert the `co2` dataset to a `tsibble` using

```
library(tsibble)
library(dplyr)

mauna <- as_tsibble(co2) |>
  # use meaningful names
  rename(year_month = index, concentration = value)
```

1. Comment on the trend, seasonality, and stationarity of the dataset using plots. Use the `mauna |> features(concentration, portmanteau_tests)` to run a portmanteau test and interpret

Linear models for time series data (`tsibble`) can be fit using the `TSLM` function from `{fable}`¹

2. Use the following code to fit a linear trend to the data and extract the fitted values and residuals. Use plots to comment on the remaining trend, seasonality, and stationarity of the residuals. Run a portmanteau test and interpret.

```
library(fable)
mauna_lm <- mauna |> model(lm = TSLM(concentration ~ trend()))
```

```
# some summaries
tidy(mauna_lm)
glance(mauna_lm)

# get fitted, residuals, ... and augment to raw data
mauna_fit <- augment(mauna_lm)
```

Note: `trend()` is a special function to generically refer to the index of the data set (i.e. `index_var(mauna) = year_month`)

Linear regression modelling

It is possible to fit multiple models simultaneously using the `model` function from the `{fable}` package. For example, we can add a linear model with polynomial trend to the data like so

```
mauna_models <- mauna |>
  model(lm_linear = TSLM(concentration ~ trend()),
        lm_quad = TSLM(concentration ~ trend() + I(trend()^2)))

tidy(mauna_models)
```

```
# A tibble: 5 × 6
  .model term estimate std.error statistic p.value
  <chr> <chr> <dbl> <dbl> <dbl> <dbl>
1 lm_linear (Intercept) 312. 0.242 1285. 0
2 lm_linear trend() 0.109 0.000896 122. 0
3 lm_quad (Intercept) 315. 0.304 1036. 0
4 lm_quad trend() 0.0674 0.00299 22.5 1.77e-76
5 lm_quad I(trend()^2) 0.0000886 0.00000618 14.3 6.86e-39
```

3. Add a new model with a cubic term. Write down the mathematical equation(s) to describe all three models.
4. Plot the residuals from the linear, quadratic and cubic fits and comment on which, if any, of these models adequately captures the trend.

Since the `mauna` tsibble includes information about the seasonality (i.e. `frequency(mauna) = 12`), we can use this information to add linear terms for seasonality to the model. This is achieved by using the `seasonal` in the `TSLM` function: `concentration ~ trend() + seasonal()`.

5. Calculate the residuals from that model and plot them. Write down the mathematical equation(s) to describe how seasonality is modeled by linear terms.
6. Calculate the ACF (correlogram) for the residual series given above and also for the first order differences of the residuals and comment on their form. How do they compare to those of the seasonally differenced data?

Other methods

7. Instead of detrending using the `index` of the time series, try differencing the data to see how the residuals behave after modelling.
8. Use the `STL` decomposition method for the `mauna` data.

Interpretation

Which of the analyses would you suggest if the goal of the analysis was to:

9. Understand the overall trend in atmospheric carbon monoxide?
10. Discern how atmospheric carbon monoxide fluctuates across a year?
11. Ascertain whether there were events that contributed to carbon monoxide fluctuations beyond any general trend and seasonal fluctuations?

References

O'Hara-Wild, Mitchell, Rob Hyndman, and Earo Wang. 2024. *Fable: Forecasting Models for Tidy Time Series*. <https://doi.org/10.32614/CRAN.package.fable>.

Footnotes

1. See [here](#) for a list of model supported by `{fable}` natively. [↗](#)